

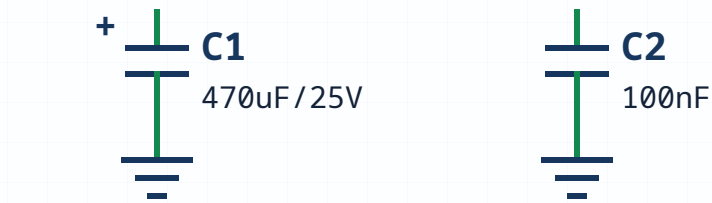
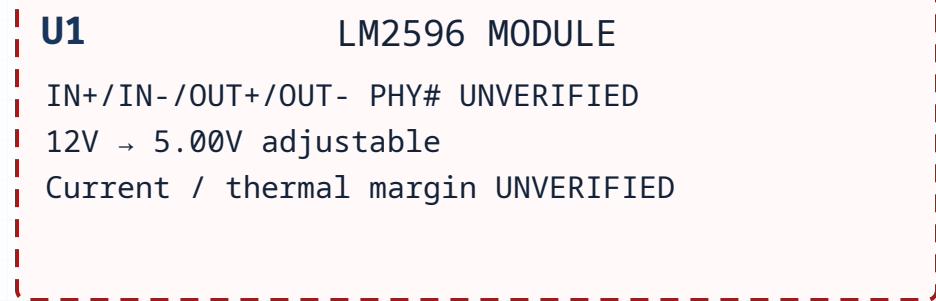
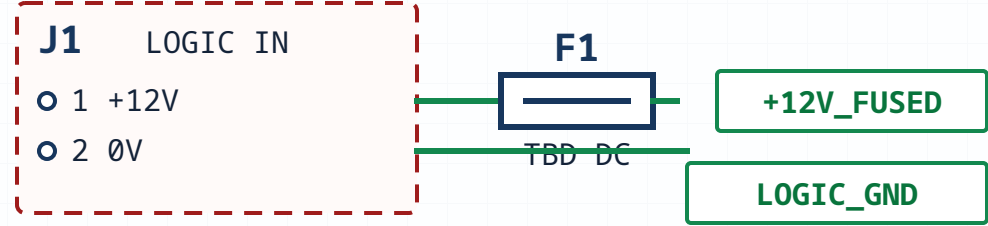
SNOWBERRY SMART GREENHOUSE – REV B CONCEPTUAL SCHEMATIC

EasyEDA-style audit canvas • exact module internals/pin numbers remain UNVERIFIED

SHEET 1 / 1

REV B0 • NOT AS-BUILT

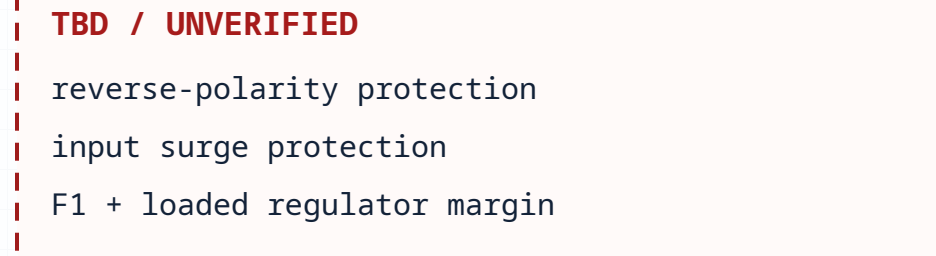
A. LOGIC POWER / REGULATION



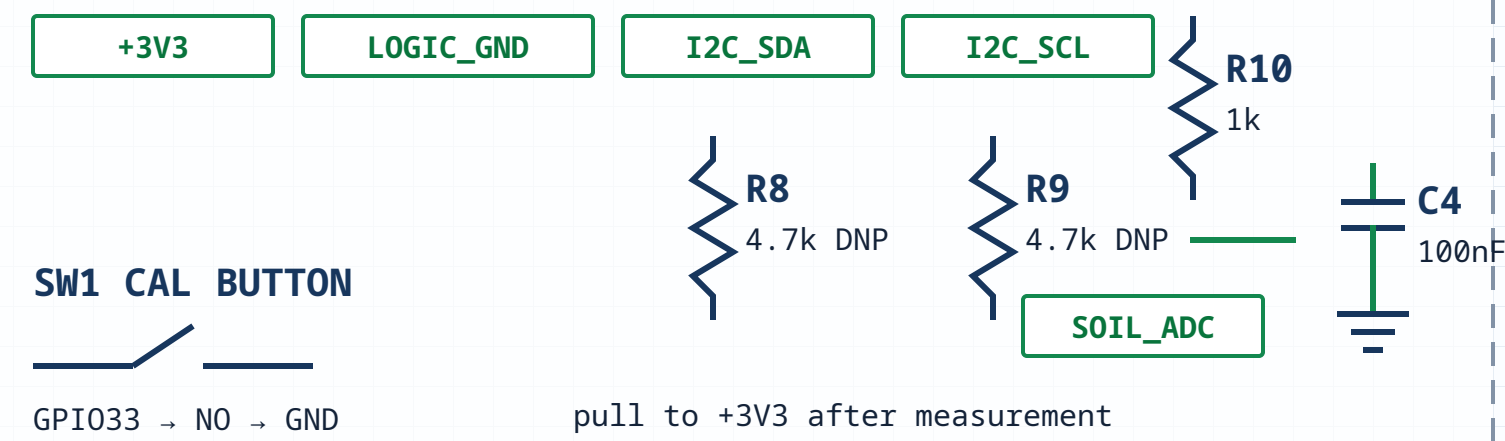
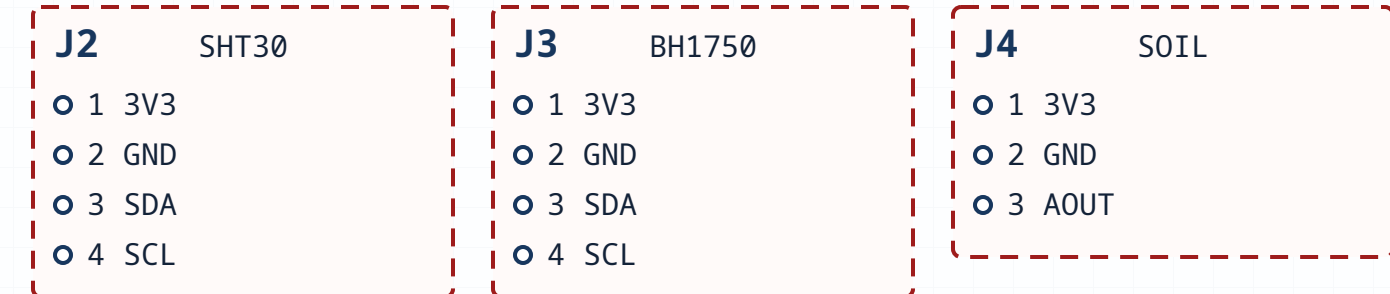
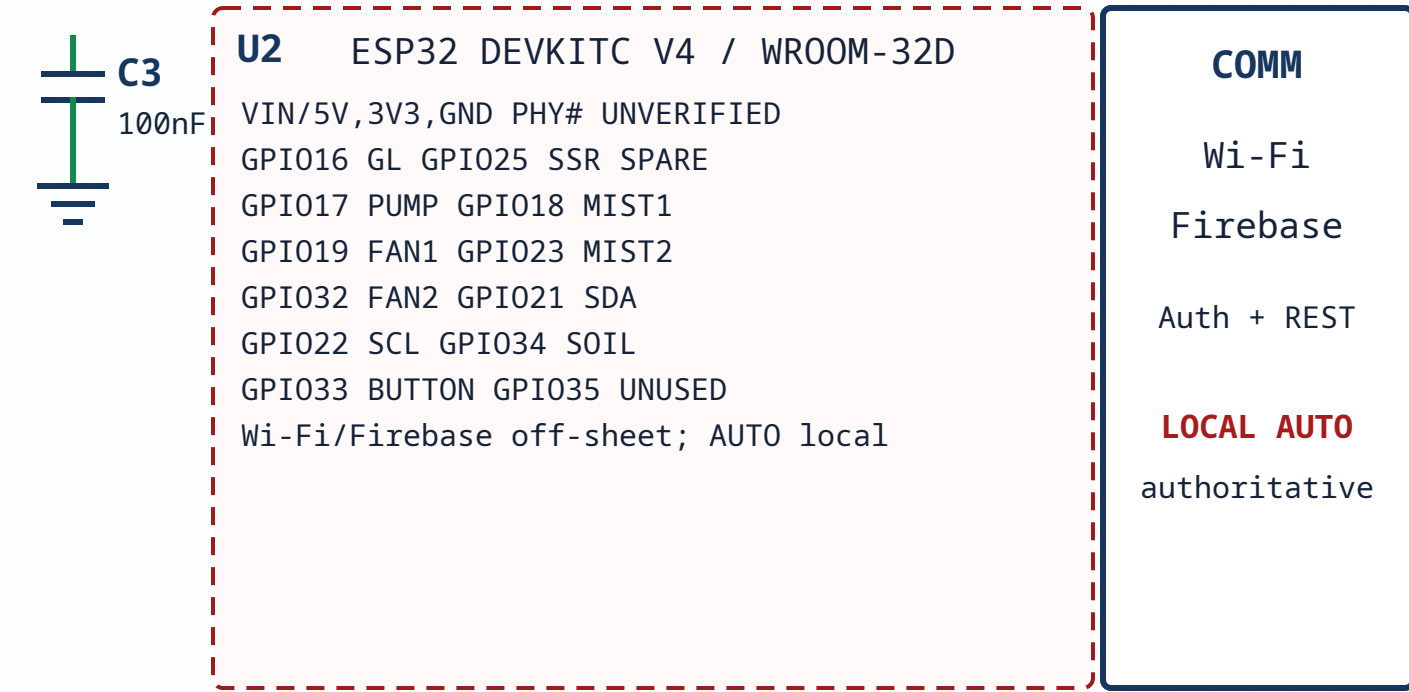
JP1 ESP POWER



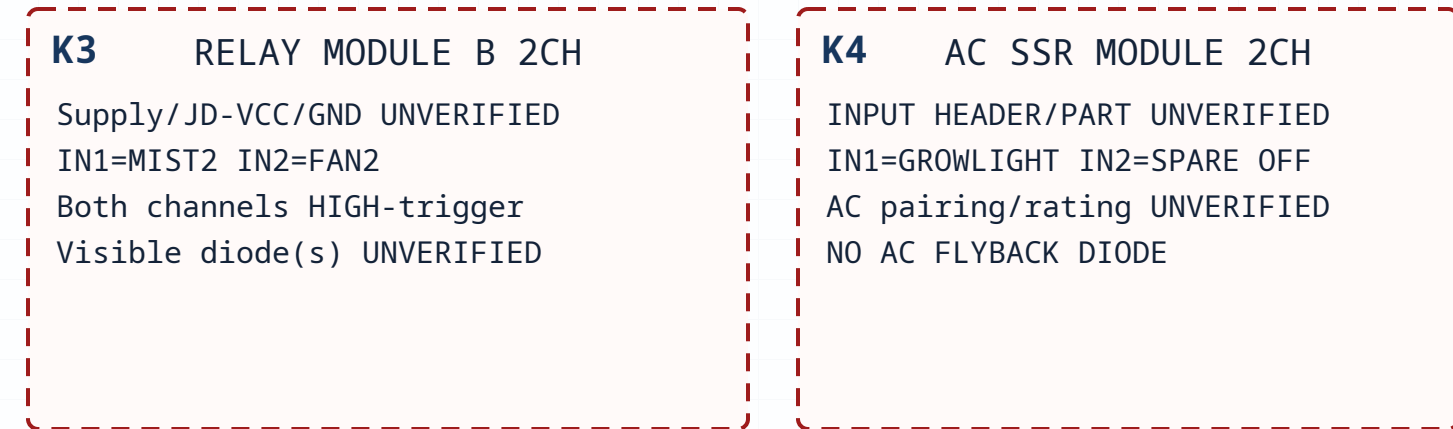
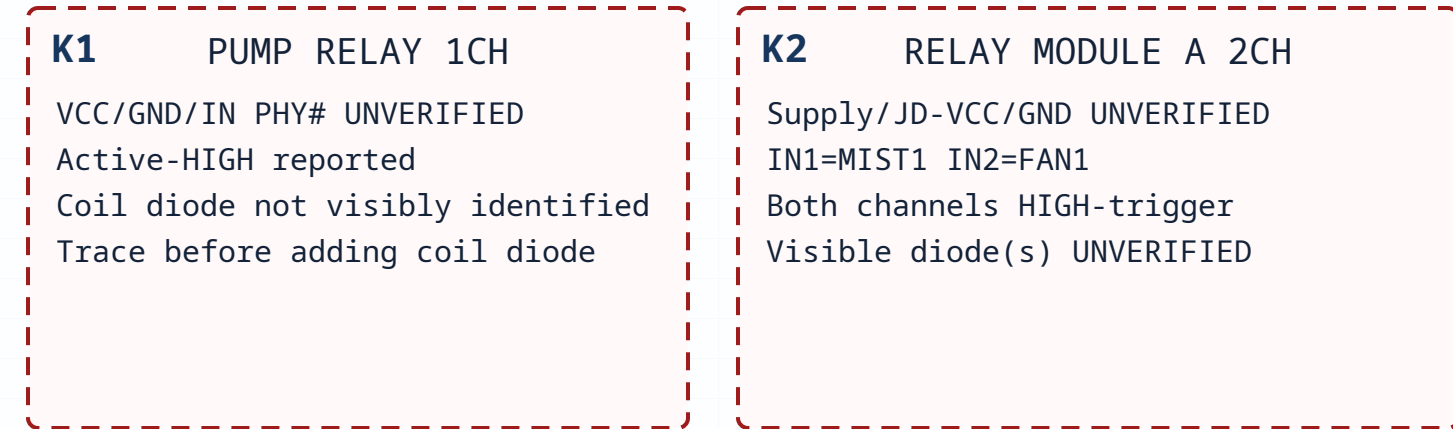
+5V_LOGIC → ESP32 VIN



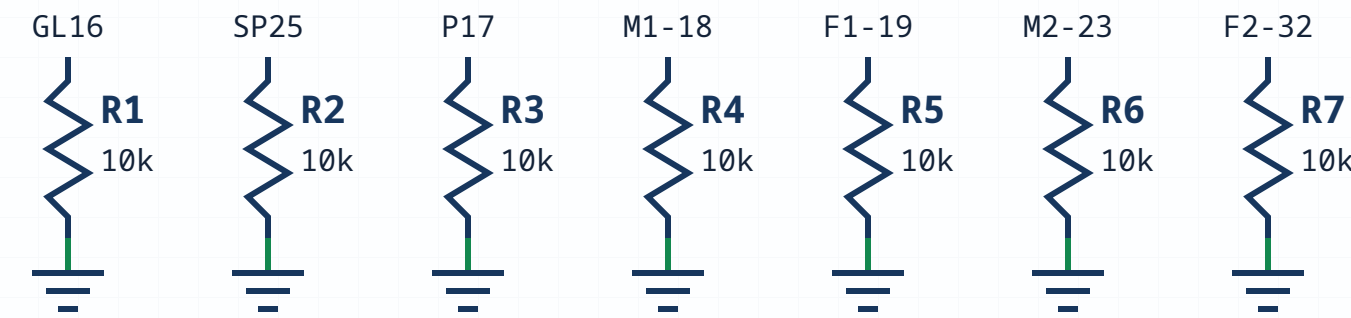
B. CONTROLLER / SENSORS



C. DRIVER / RELAY CONTROL



10k ACTIVE-HIGH PULLDOWNS

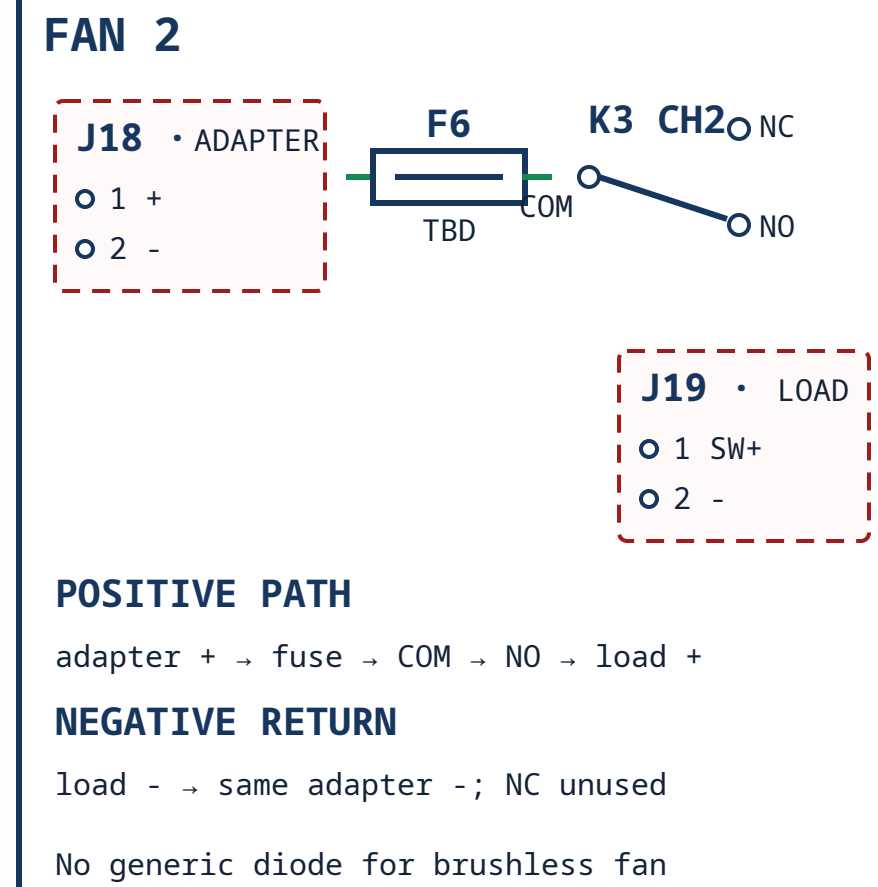
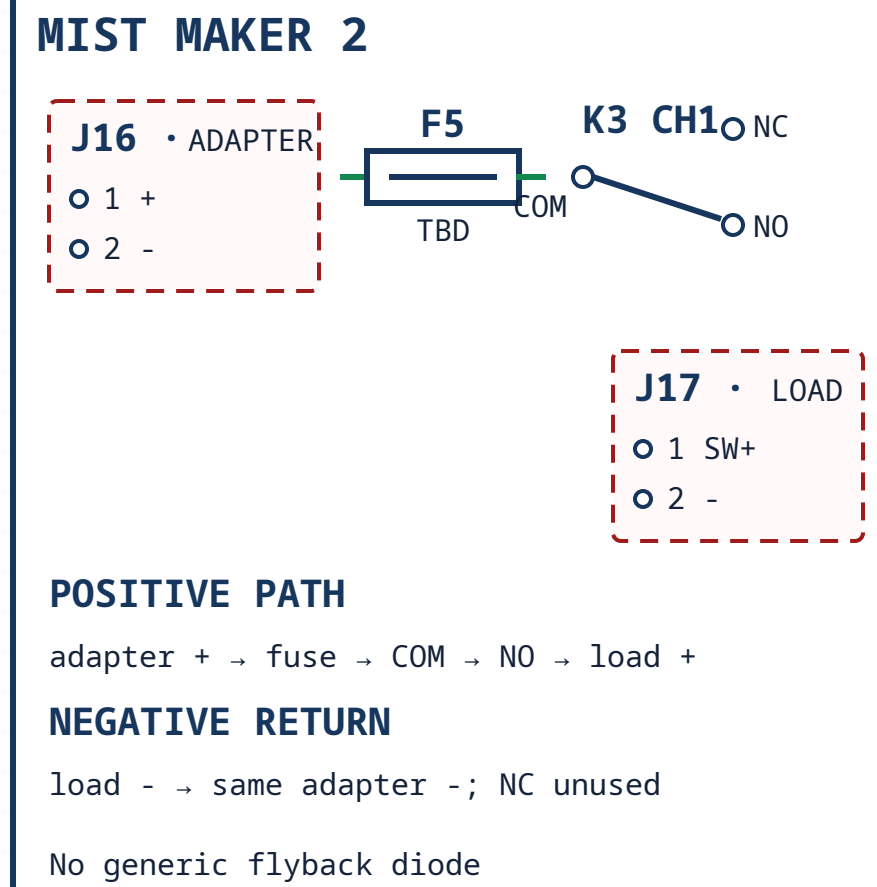
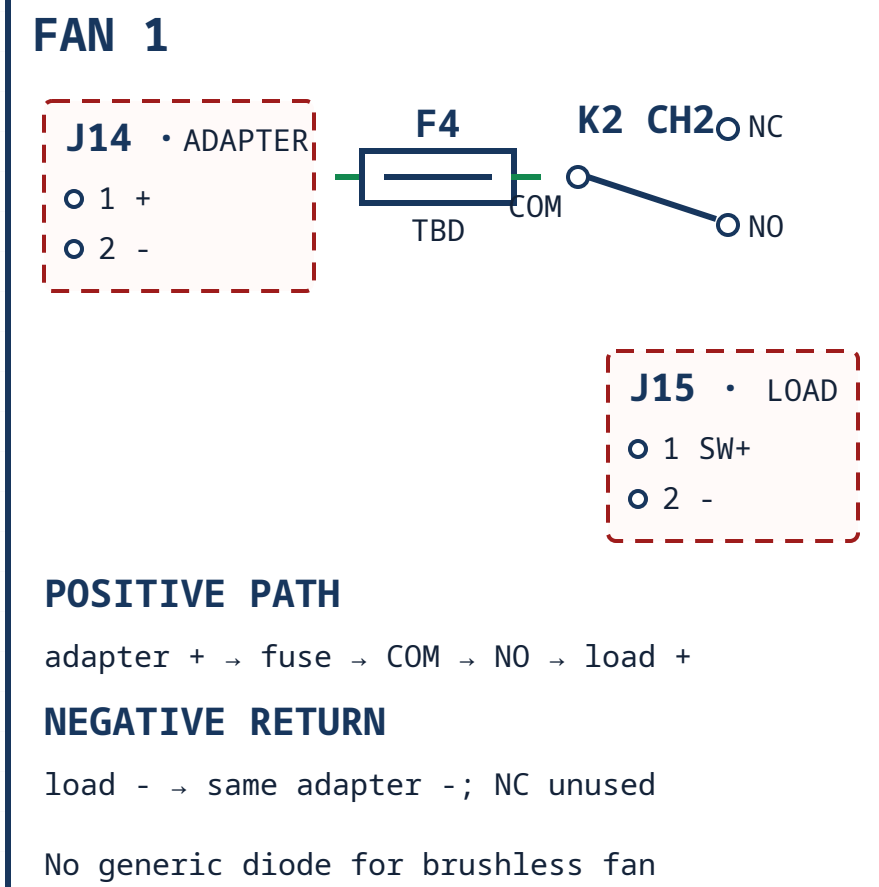
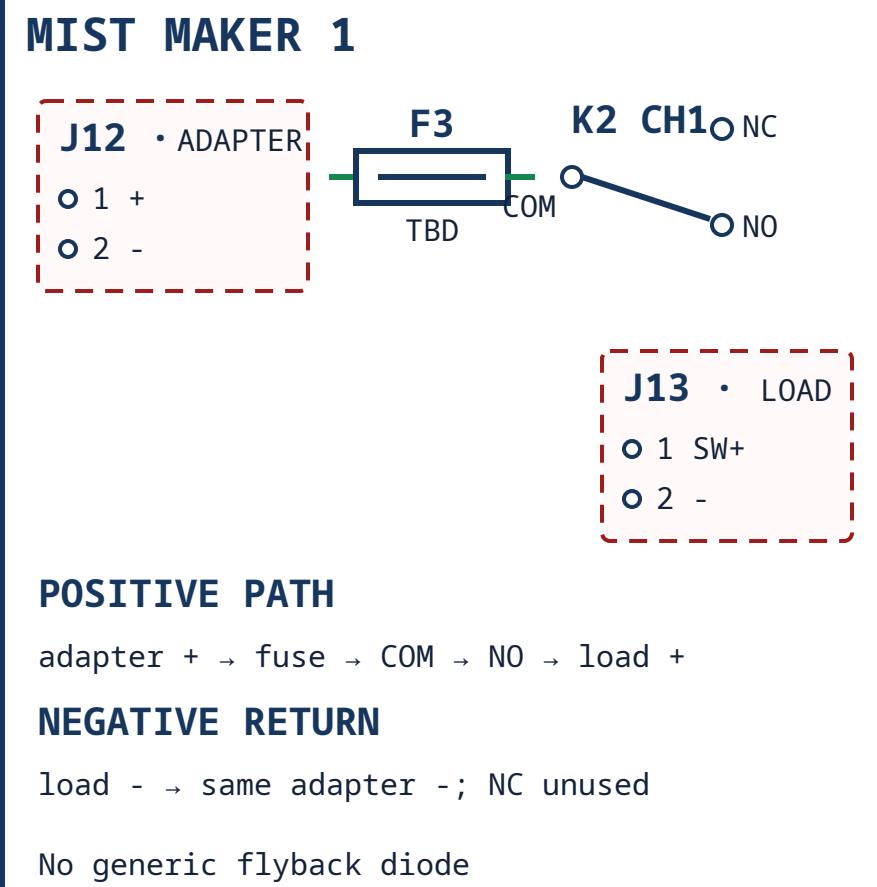
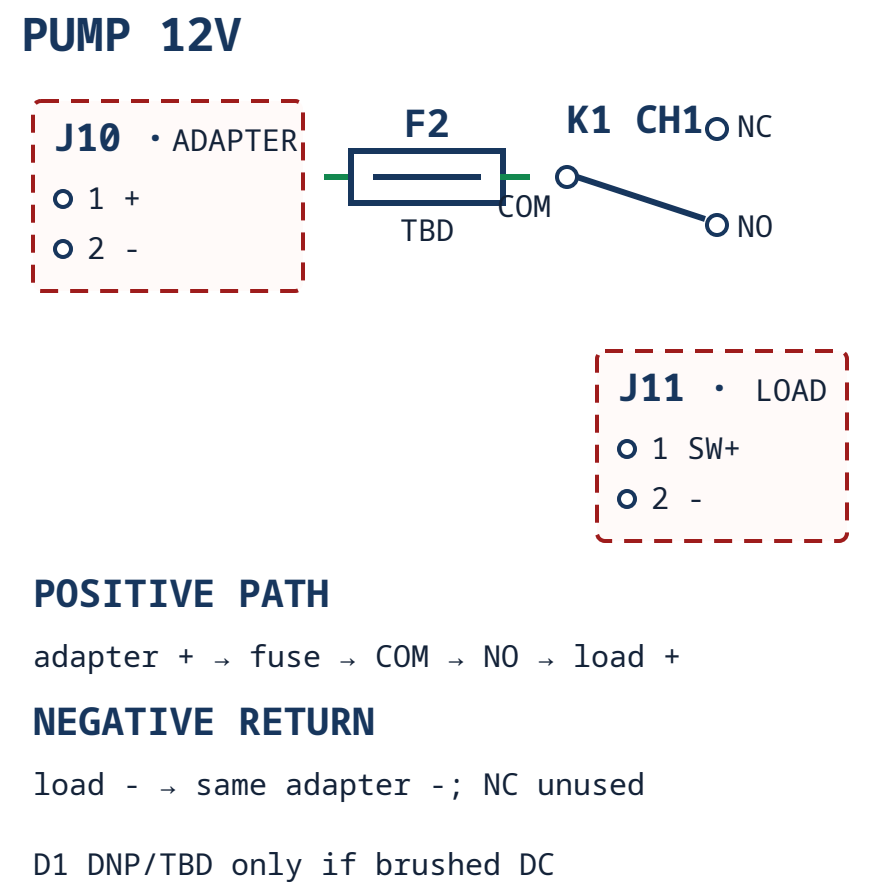


GPIO LOW=OFF • GPIO HIGH=ON

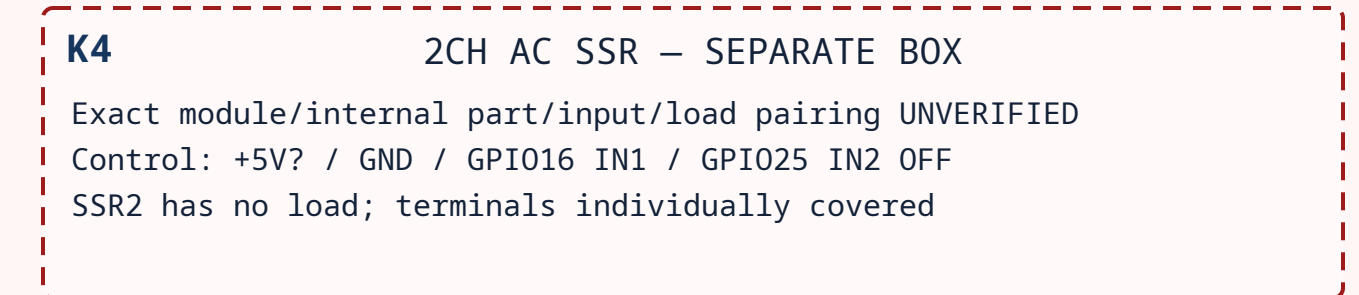
Verify all 5 relay + 2 SSR inputs with pulldowns

If marginal: add verified non-inverting driver

D. ISOLATED DC LOAD DOMAINS



E. SEPARATE AC SSR BOX



ISOLATION BARRIER RATING UNVERIFIED



Named nets avoid ambiguous crossings:



J21 GROWLIGHT OUT

- 1 L_SW
- 2 N
- 3 PE

CONNECTIONS

L → F7 → K4 CH1 → L_SW

N → N direct

PE → PE direct where required

Switch verified LIVE only

SSR OFF ≠ service isolation

No AC on matrix board

AC ITEMS UNVERIFIED / TBD

- exact SSR part, load category, leakage, thermal rise
- F7 value/type/voltage/breaking capacity
- L/N/PE identification and equipment class
- cable, connector and enclosure ratings
- upstream disconnect and residual-current protection

TEST POINTS / COMMUNICATION

TP1 GND • TP2 +12V • TP3 +5V • TP4 +3V3

TP5-TP11 outputs • TP12 SDA • TP13 SCL • TP14 SOIL

Wi-Fi/Firebase off-sheet; local AUTO remains authoritative

CONCEPTUAL SCHEMATIC – Verify all pinouts, ratings, protection and physical wiring before fabrication or deployment.